## 📘 MODULE 3: UNDERSTANDING SPLUNK DATA

### 3.1 Core terms

| Term | Meaning | Example |
|---|---|---|
| **Event** | One record of data, usually one log line, with a timestamp | Oct 7 10:15:01 web1 sshd[812]: Failed password for root from 10.0.0.5 |
| **Raw event** (_raw) | The original text exactly as received | The whole line above |
| **Field** | A named piece of information inside or about an event | user, src_ip, host |
| **Field value** | The actual data held by the field | root, 10.0.0.5 |
| **Timestamp** (_time) | When the event happened | 2026-10-07 10:15:01 |
| **Metadata** | Data *about* the data: host, source, sourcetype, index | |

**Field vs value:** the field is the *label*, the value is the *content*.
user=root → field user, value root.

**Event vs field:** an event is the whole record. Fields are pieces extracted from it.

### 3.2 The four default metadata fields (memorize)

| Field | Answers | Example |
|---|---|---|
| **host** | *Which machine* generated the data? | web1, ubuntu-lab |
| **source** | *Where exactly* did it come from? (file path, port, URL) | /var/log/auth.log |
| **sourcetype** | *What format/kind of data* is it? | linux_secure, access_combined |
| **index** | *Where is it stored*? | main, web |

Also: _time, _raw, _indextime (when Splunk indexed it).

### 3.3 The comparisons you must master

**HOST vs SOURCE**
Host is the **machine**. Source is the **file or input** on it.
host=web1 and source=/var/log/apache2/access.log.

**SOURCE vs SOURCETYPE**
Source is **where** it came from. Sourcetype is **what format** it is.
Many sources can share one sourcetype: /var/log/apache2/access.log on 10 servers are all access_combined, though each has its own host. Sourcetype drives parsing, timestamps, and field extraction, so it matters more than source.

**INDEX vs SOURCETYPE**
Index is **storage and access control**. Sourcetype is **format**. One index can hold many sourcetypes (index=main may have both Linux and web data).

**Correcting your Q5 answer:** sourcetype is not the operating system, and host is not the person. Host is the machine name.

**3.4 Tags and event types (brief)**

- **Event type:** a *saved search* that labels events. Example: eventtype=failed_login = sourcetype=linux_secure "Failed password".
- **Tag:** a label attached to a field value, such as tag=authentication. Tags power CIM mapping (Module 17).

**3.5 Example searches**

**Query**

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index=main sourcetype=linux_secure host=ubuntu-lab "Failed password"

**What it does:** shows failed-password events from one host and one data type.
**Line-by-line:** index=main pick the storage location · sourcetype= pick the format · host= pick the machine · "Failed password" quoted text matches that exact phrase in _raw.
**Expected result:** a list of events with the phrase highlighted.
**Common mistakes:** forgetting quotes around multi-word phrases, or using the wrong time range.
**Interview explanation:** "I narrow by index, sourcetype, and host first so Splunk reads less data, then filter on keywords."

**3.6 Hands-on**

1. Run index=main | head 20. Click an event and expand it.
2. Write down the host, source, sourcetype, index, and _time of one event.
3. Click sourcetype in the left sidebar. How many distinct values do you see?

### 3.7 Interview Q&A

- **Source vs sourcetype?** Source is the origin (path/port). Sourcetype is the data format that controls parsing.
- **Host vs source?** Host is the machine, source is the file or input.
- **What are the default fields?** host, source, sourcetype, index, _time, _raw.
- **Event vs field?** An event is the full record, a field is a named value extracted from it.

### 3.8 Revision

Event = one record · field = label, value = content · host=machine · source=origin · sourcetype=format · index=storage.